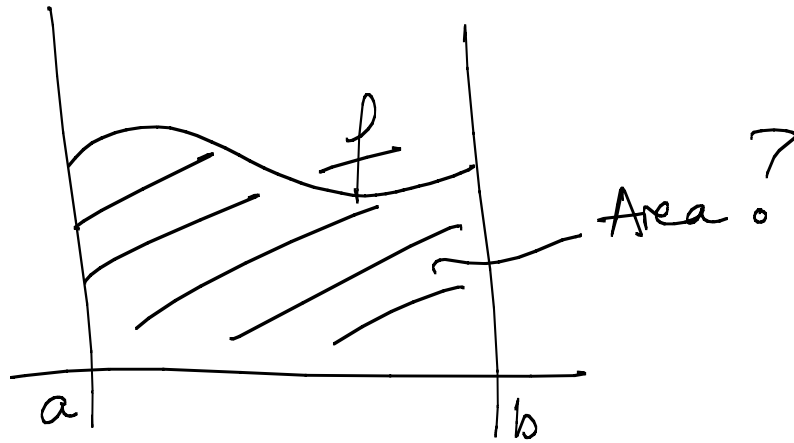


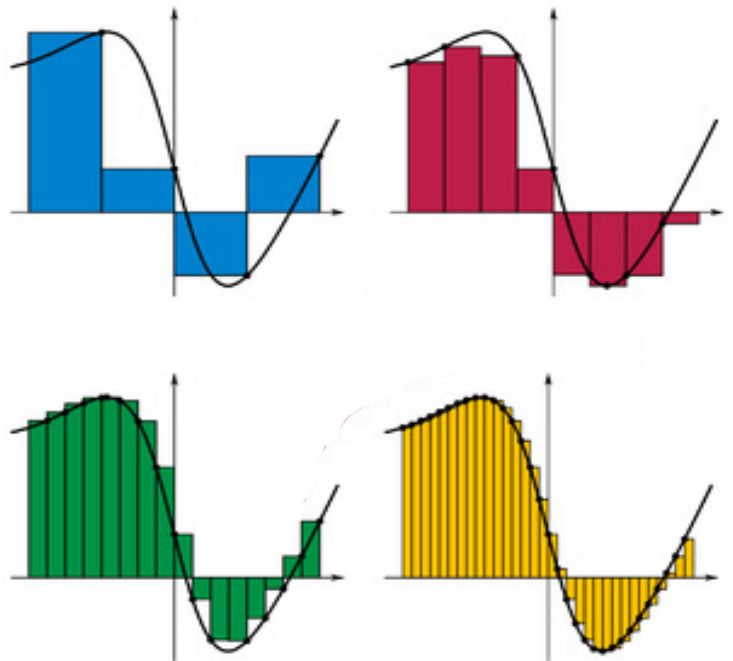
Problem:



If f is continuous on $[a, b]$ and $f(x) > 0$ on $[a, b]$ how do you calculate (or estimate) the area between the graph, the x -axis and the lines $x=a$ and $x=b$?

Riemann-sums

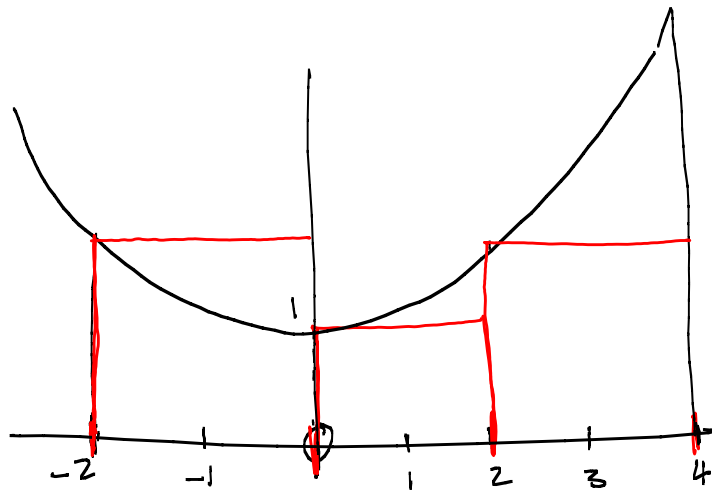
- Left-hand sum
- Right-hand sum
- Over estimate
- Under estimate.



Left-hand Riemann sum

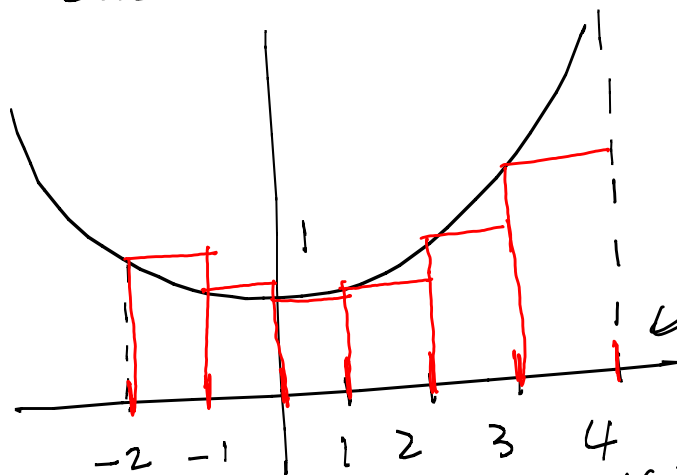
Example 1.

(1) $f(x) = x^2 + 1$ on $[-2, 4]$ using 3 subintervals.



$$\begin{aligned}\text{Approx area} &= 2 \times f(-2) + 2 \times f(0) + 2 \times f(2) \\ &= 2 \times 5 + 2 \times 1 + 2 \times 5 = \underline{\underline{22}}\end{aligned}$$

(2) Using 6 subintervals



Left-hand sum
 $f(4)$ is not
used

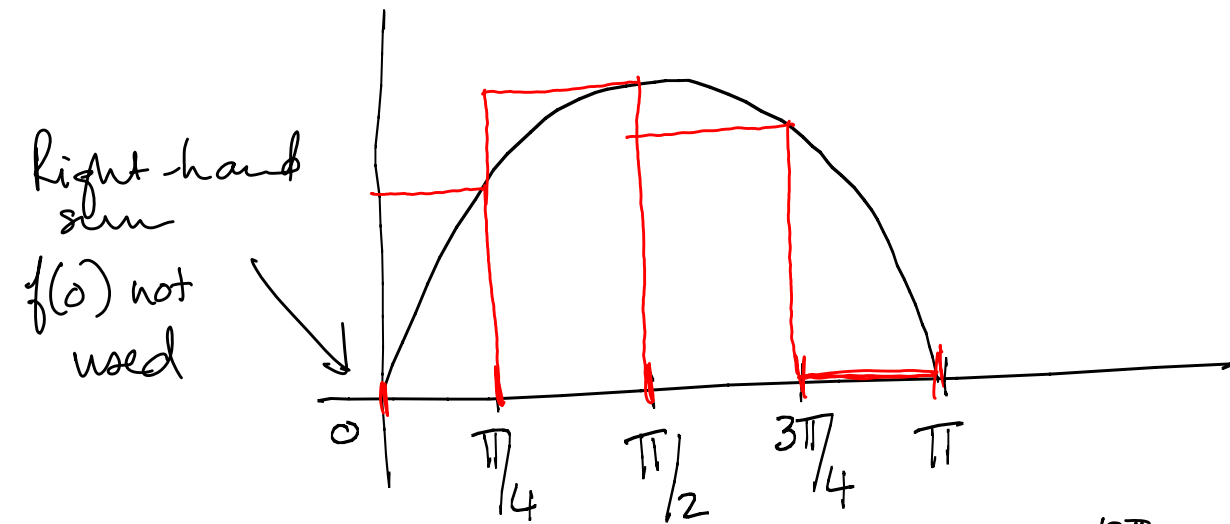
$$\begin{aligned}\text{Approx area: } &f(-2) + f(-1) + f(0) + f(1) + f(2) + f(3) \\ &= 28.\end{aligned}$$

28

Right-hand Riemann-sum

Example

(1) $f(x) = \sin x$ on $[0, \pi]$ using 4 rectangles

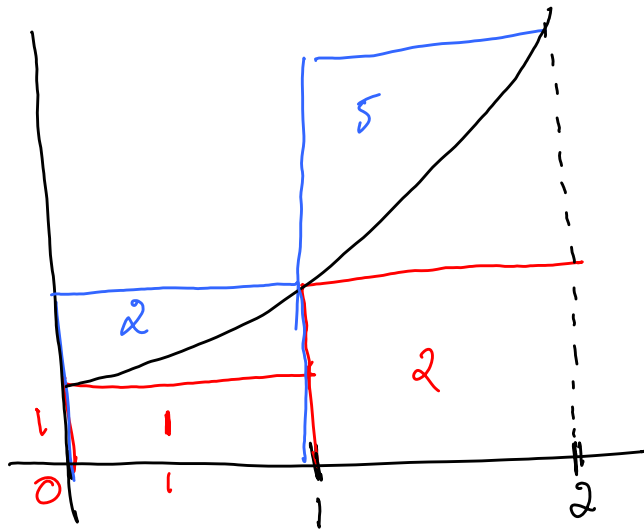


$$\begin{aligned}\text{Approx area: } & \frac{\pi}{4} \left[f\left(\frac{\pi}{4}\right) + f\left(\frac{\pi}{2}\right) + f\left(\frac{3\pi}{4}\right) + f(\pi) \right] \\ &= \frac{\pi}{4} \left[\frac{1}{\sqrt{2}} + 1 + \frac{1}{\sqrt{2}} + 0 \right] \\ &= \frac{\pi}{4} \left[\frac{2}{\sqrt{2}} + 1 \right]\end{aligned}$$

Over estimate and under estimate

① $f(x) = x^2 + 1$ on $[0, 2]$ using two rectangles

f is increasing
on the
interval



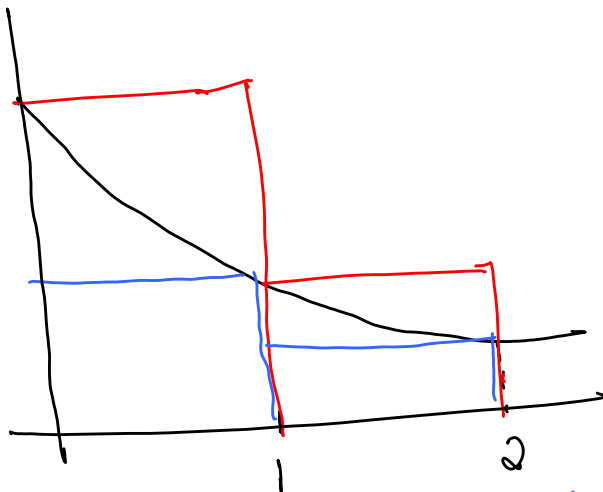
Real area:
 $\frac{14}{3}$

Left-hand Riemann sum
Underestimate (3)

Right-hand Riemann sum
Over estimate (7)

② $f(x) = e^{-x}$ on $[0, 2]$ using 2 rectangles

f is decreasing
on the interval



Left-hand Riemann sum
Over estimate

Right-hand Riemann sum
Under estimate